

Final Design Report

Micromouse X Subsystem



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Declaration

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Contents

1	Introduction	1
1.1	Problem Description	1
1.2	Scope and Limitations	1
1.3	GitHub Link	1
2	Requirements Analysis	2
2.1	Traceability Analysis	2
2.2	Requirements	2
2.3	Specifications	2
2.4	Acceptance Testing Procedures	3
3	Subsystem Design	5
3.1	Design Decisions	5
3.1.1	Charging Solution Selection	5
3.1.2	BQ24074RGT Charging Current Switching Mechanism	5
3.1.3	USB Power Delivery Controller Selection	6
3.2	Final Design	7
3.3	Bill of Materials	9
3.4	Failure Management	9
3.5	System Integration and Interfacing	9
4	Discussion and Results	11
4.1	AT01: USB Charging Verification	11
4.2	AT02: Fast Charge triggers	11
4.3	AT03: Standard Charging Current	12
4.4	AT04: Fast Charging Current	13
4.5	AT05: USB-C 9 V Output Verification	13
5	Conclusion	14
5.1	Recommendations	14
6	Reflection	15

Chapter 1

Introduction

1.1 Problem Description

This project is derived from the Micro-Mouse competition hosted all over the world. A Micro-Mouse is a miniature autonomous robot designed to navigate a maze and reach its centre in the shortest amount of time. It utilises sensors, motors and a processing unit to map out the maze, optimise the quickest route and race to the finish line.

The objective of this project is to design a subsystem to provide a host of power requirements to a Micro-Mouse. In particular, the subsystem is required to allow for motor control, external switching, and host of battery related tasks. Students are expected to draw on their knowledge over their first two years of study and work as a team to design the power subsystem.

1.2 Scope and Limitations

The project entails only the design of the power subsystem of the Micro-Mouse on a printed circuit board, omitting the designs of the sensing and processing units and the construction of the Micro-Mouse itself.

This report entails 4 specific objectives that the power subsystem is to meet, namely:

- Utilise an INA219 to monitor battery status through the I2C interface.
- Integrate the USB Type-C standard.
- Have the battery charge at 9 V from the input pin on the JST header.
- Have two charging currents dependent on speed of charge required, one at 200 mA and another at 600 mA.

Additionally, the power subsystem must be within limits outlined by the brief. The power system is required to interface with the control module of the Micro-Mouse using a 2x16 JST pin header, must adhere to a strict budget of \$70, the dimensions of the PCB must be within 80x62 mm and components may only be sourced from JLCPCB.

1.3 GitHub Link

The full design and supporting materials can be found in the following GitHub repository.

<https://github.com/segcboi/EEE3088F>

Chapter 2

Requirements Analysis

2.1 Traceability Analysis

Table 2.1: Requirements Traceability Matrix

#	Requirements	Specifications	Acceptance Test
1	RQ01	SP01	
2	RQ01	SP02	
3	RQ01	SP03	
4	RQ02	SP04	AT01
5	RQ03	SP05	AT02
6	RQ03	SP06	AT03
7	RQ03	SP07	AT04
8	RQ04	SP08	AT05
9	RQ05	SP09	

2.2 Requirements

The requirements for a Micro-Mouse power module are described in [Table 2.2](#).

Table 2.2: Requirements of the power subsystem.

Requirement ID	Description
RQ01	Utilize an INA219 connected to the I2C bus to monitor the battery
RQ02	Charge the Li-Po battery
RQ03	Have two charging modes, one standard and another faster
RQ04	Negotiate a high voltage out of the USB host
RQ05	Integrate USB type C
RQ06	Provide a 3.3 V with 5% accuracy
RQ07	Provide a 5 V V with 5% accuracy
RQ08	Have a switch that disables 3V3 and 5V outputs
RQ09	Operate four motors bidirectionally
RQ10	Provide two external load switches

2.3 Specifications

The specifications, refined from the requirements in [Table 2.2](#), for the Micro-Mouse power module are described in [Table 2.3](#).

Table 2.3: Specifications of the sensing subsystem derived from the requirements in Table 2.2.

Specification ID	Description
SP01	Configure the INA219 pin A0 and A1 to GND and Power respectively.
SP02	INA219 placement between the battery and the load
SP03	INA219 connects to the I2C bus
SP04	9V output from the USB-C feeds into the charging IC
SP05	Charging mode must be controlled through the logic of the Fast Charge pin
SP06	Lower charging mode charges at 200 mA
SP07	Higher charging mode charges at 600 ± 100 mA
SP08	Utilize and configure a USB Power Delivery (PD) controller to negotiate 9V out of the Type C port.
SP09	USB-C port integrated into the power subsystem

2.4 Acceptance Testing Procedures

Table 2.4: Subsystem acceptance tests

ID	Description	Testing Procedure	Pass/Fail Criteria
AT01	Confirm that the 9 V output of the USB charges the battery.	- Allow the battery to fully discharge and measure the voltage. - Plug the subsystem into a power source supporting USB Power Delivery protocol. - Wait 10 minutes to allow charging - Disconnect the battery and measure the voltage again.	Pass if the voltage of the battery is higher than before after allowed to charge, fail if otherwise.
AT02	Confirm that fast charging can be triggered using the Fast charge pin	- Connect a voltmeter to the Fast Charge logic pin. - Toggle the pin between logic HIGH and logic LOW. - Observe the corresponding charging current in each state using a multi-meter or current probe.	Pass if a high on the logic pin results in a high charge current and if a low on the logic pin results in a low current, fail otherwise
<i>Continued on next page</i>			

ID	Description	Testing Procedure	Pass/Fail Criteria
AT03	Verify if the standard charging mode operates	- Confirm that the fast-charging logic pin is set to LOW. - Confirm that the current going into the battery is 200 mA.	Pass if the charging current is at 200 ± 10 mA, fail if otherwise
AT04	Verify if the fast charging mode operates	- Confirm that the fast-charging pin is set to logic HIGH. - Confirm that the current going into the battery remains between 500 mA and 700 mA continuously for 10 seconds.	Pass if the charging current is at 600 ± 100 mA, fail if otherwise
AT05	Confirm if the USB-C port outputs 9V	- Plug the system into a power source that supports USB Power Delivery protocol. - Use a multimeter to measure the voltage at the USB-C output line. - Confirm that the PD controller negotiates and outputs 9 V.	Pass if the output of the USB is at 9 ± 0.15 V, fail if otherwise

Chapter 3

Subsystem Design

3.1 Design Decisions

Three key design considerations are highlighted below:

3.1.1 Charging Solution Selection

To meet the requirements and specifications of the charging requirements that the power subsystem is capable of, a system to set charging levels based on logic from the Micro Mouse motherboard is required. This design links back to Requirement RQ03, which concerns the charging speed of the battery at different logic states of the Fast Charge pins, and specifications SP05, SP06 and SP07, which dictate the mode of charging and charging speed. Many charging ICs are available to achieve this requirement and specifications, and the two final candidates' properties are highlighted below.

Aspect	BQ24295	BQ24074RGT RGT
Charge Current Control	Programmable using I2C	Fixed via external resistors
Charging mode selection	Programmable using I2C	External hardware switching
Complexity of implementation	More complex to implement, requiring software and external components to configure correctly	Simple to implement on the subsystem using external components
System Integration	Requires I2C host to implement	Simple set up of IC on board

Table 3.1: Comparison between BQ24295 and BQ24074RGT

The BQ24074RGT is ultimately chosen due to its simplicity of implementation. Using the BQ24074RGT produces a current output set by external resistors. This allows the use of a switching mechanic such as an BJT or MOSFET to change between a series and parallel network of resistors to achieve the two specified currents. This can be verified with acceptance tests AT03 and AT04 that the value of the output current of the BQ24074RGT is correct in the two charging modes.

3.1.2 BQ24074RGT Charging Current Switching Mechanism

As stated above, the BQ24074RGT can be configured to output certain currents by the effective resistance that controls the output current value. In order to do this, a switching circuit needs to be utilised using a transistor to toggle between different resistor network topologies.

The charging current of the BQ24074RGT is set by choosing an appropriate value for the resistor R_{SET} to set the output current, I_{SET} . The relationship is defined in the formula sheet as,

$$I_{SET} = \frac{K_{ISET}}{R_{SET}}$$

where K_{SET} is a constant defined in the formula sheet as $890 \Omega \cdot \text{A}$. To produce charging currents of 200 mA and 600 mA, resistor values of 4450Ω and 1483Ω are required, respectively.

The transistor choice was between a BJT and MOSFET, specifically the MMBT5551L and the IRFR024NTRPB.

Aspect	MMBT5551L	IRFR024NTRPB
Control Type	Current driven	Voltage driven
Application	Low-power switching	High-power switching
Footprint and Package	Small, SOT-23 package	Larger, DPAK package
Cost	\$0.0245	\$0.237
Availability	Readily available	Less commonly stocked

Table 3.2: Comparison between MMBT5551L and IRFR024NTRPB

The MMBT5551L is ultimately chosen for its simplicity of operation and lower power requirements, especially since the value of the $R_{DS(on)}$ at the Fast Charge pin is not known. Using this transistor allows the use of logic levels from the Fast Charge pin to trigger switching between the two resistor topologies that will bias the BQ24074RGT RGT to output the two required charging currents, satisfying RQ03 is verified through AT02.

The final switching mechanism is pictured in Figure 3.1.

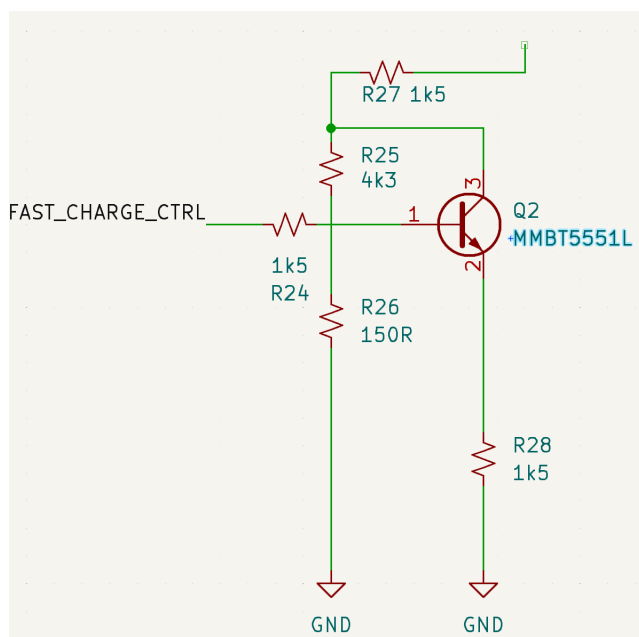


Figure 3.1: Switching mechanism to set charge currents

3.1.3 USB Power Delivery Controller Selection

To meet the USB PD requirements and safely negotiate 9 V from a USB-C host, a compliant PD controller is needed. Linking back to requirement RQ04, which concerns negotiating 9 V from the USB host and specification SP08, which concerns the utilisation and configuration of USB PD controller,

a multitude of ICs could be considered to achieve these this, with the HUSB237 and STUSB4500 highlighted below as final candidates for the choice of USB PD controller.

Aspect	HUSB237	STUSB4500
PD Profile Configuration	Fixed profiles	Programmable profiles
Firmware Requirements	None	Programmable via I2C
Autonomous Operation	Fully autonomous plug and play operation	Needs I2C configuration on startup
Complexity of Implementation	Simple to implement on the sub-system	Complex to implement, requiring software configuration
Cost	\$0.1869	\$1.4085

Table 3.3: Comparison between HUSB237 and STUSB4500

The HUSB237 is ultimately chosen, due mainly to it's cost-effectiveness, simplicity, autonomous negotiation and hardware based integration. The HUSB237 satisfies SP08 with minimal software overhead, making it ideal for a hardware based approach of negotiating 9 V out of the host as required by RQ04. This can be verified with acceptance test AT05 that the correct voltage is achieved.

3.2 Final Design

The final design uses the **BQ24074RGT RGT** charger IC, controlled via external resistors to set charging currents at **200 mA** and **600 mA** through a transistor switching mechanism. A **MMBT5551L BJT** is used for the switching circuit, chosen for its simplicity and reliability.. This satisfies Requirement **RQ03** and specifications **SP05**, **SP06** and **SP07**, verified by **AT02**, **AT03** and **AT04**.

The **HUSB237 USB PD controller** negotiates 9V from a USB-C host autonomously, meeting **RQ04** and **SP08** with minimal software overhead, verified by **AT05**.

This design balances simplicity, cost, and functionality, providing efficient charging with easy integration.

3.2. Final Design

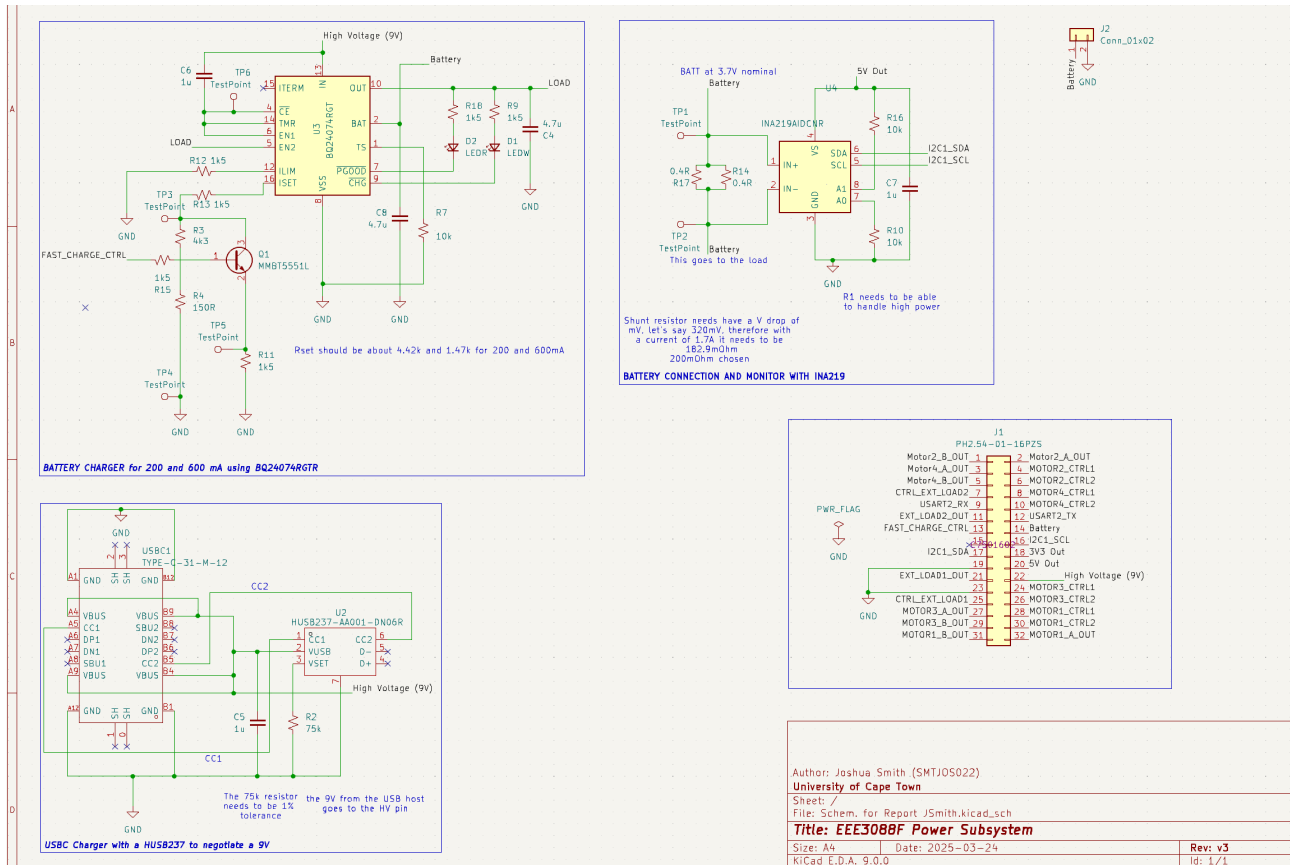
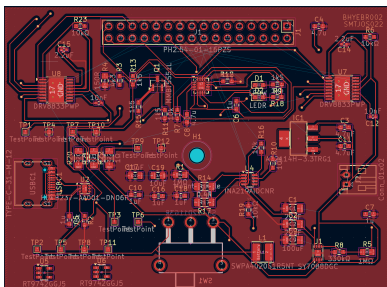
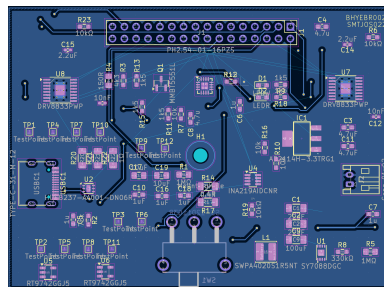


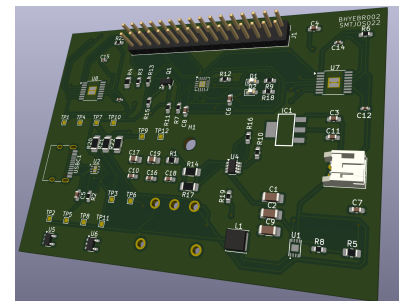
Figure 3.2: Schematic



(a) Front PCB



(b) Back PCB



(c) 3D PCB

Figure 3.3: PCB

3.3 Bill of Materials

Designator	Qty.	JLCPCB Part No.	Description	Price
C4, C8	2	C19666	4.7F Capacitor	\$0.0904
C5, C6, C7	3	C15849	1F Capacitor	\$ 0.0612
D1	1	C2290	White 0603 LED	\$0.0204
D2	1	C2286	Red 0603 LED	\$0.0112
R2	1	C23242	75k Resistor	\$0.0022
R3	1	C23159	4.3k Resistor	\$0.0220
R4	1	C17471	150 Resistor	\$0.0038
R9, R11, R12, R13, R15, R18	6	C22843	1k5 Resistor	\$0.0132
R14, R17	2	C333100	400m Resistor	\$0.1540
R7, R10, R16	3	C25804	10k Resistor	\$0.012
Q1	1	C41095	NPN BJT	\$0.245
U2	1	C22373734	HUSB237	\$ 1.1214
U3	1	C54313	BQ24074RGT RGT	\$ 1.4490
U4	1	C87469	INA219AIDCNR	\$ 1.149
USBC1	1	C165948	USB-C Receptacle	\$0.3394
Total				\$4.0296

Table 3.4: Bill of materials

3.4 Failure Management

Name	Description
Test Points	The use of test points on the PCB allows passive components to be removed and replaced using hand soldering techniques.
LED Indicator	Provides visual feedback for power presence, charging status, or fault conditions, aiding in early fault detection on the charging system.
Jumpers	In hindsight, jumpers in strategic locations along the board would have been beneficial in bypassing connections between failed components.

Table 3.5: Failure Management Strategies in PCB Design

3.5 System Integration and Interfacing

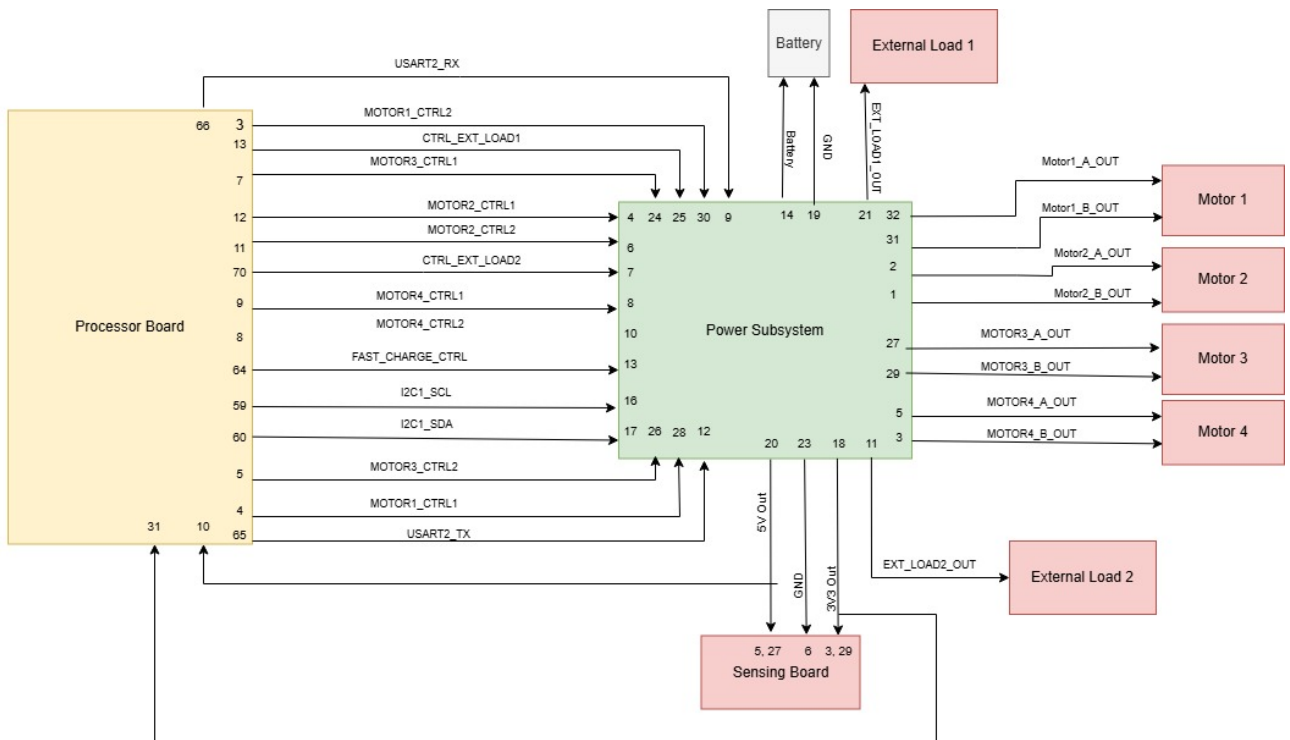


Figure 3.4: Interface Diagram

Chapter 4

Discussion and Results

4.1 AT01: USB Charging Verification

The objective of AT01 was to verify that a 9 V input into the BQ24074RGT would enable battery charging. Initially, this test procedure involved allowing a discharged battery to charge and then measuring its voltage after 10 minutes. However, this approach was not feasible due to the unavailability of the battery during testing.

Instead, a resistor bank was constructed using five $100\,\Omega$ resistors connected in parallel, resulting in an effective resistance of $20\,\Omega$. This parallel configuration ensured that less current flowed through each resistor, reducing power dissipation per resistor and protecting them from thermal damage.

The resistor bank was then connected across the battery output pins to simulate a load. Charging was initiated by supplying 9 V to the BQ24074RGT. The voltage across the dummy load was measured to be 1.66 V, which allows calculation of the charging current using Ohm's Law:

$$I_{\text{CHG}} = \frac{V_{\text{BATT}}}{R_{\text{DUMMY}}} = \frac{1.66}{20} = 83\text{ mA}$$

This confirms that the charger is sourcing current into the load using a 9 V input, thereby passing AT01. However, the measured current is significantly lower than the expected value of approximately 200 mA, which is the standard charging current configured via the ISET resistor. This discrepancy suggests that further investigation is needed to confirm whether there are potential issues with the charging IC circuitry.

4.2 AT02: Fast Charge triggers

The objective of AT02 was to verify if logic signal of the FAST_CHARGE_CTRL pin would result in a higher battery charge current from the BQ24074RGT. Using the same $20\,\Omega$ dummy load as AT01, measurements were taken with the FAST_CHARGE_CTRL pin at both low (0V) and high (3.3V) states.

Standard charge mode (0V): 1.66V across load, resulting in 83 mA.

Fast charge mode (3.3V): 1.69V across load, resulting in 84.5 mA.

The negligible current increase of 1.5 mA falls significantly short of the expected difference between standard (200 mA) and fast charge (600 mA) modes. This minimal response suggests the BQ24074RGT is operating in a current-limiting mode regardless of the FAST_CHARGE_CTRL input.

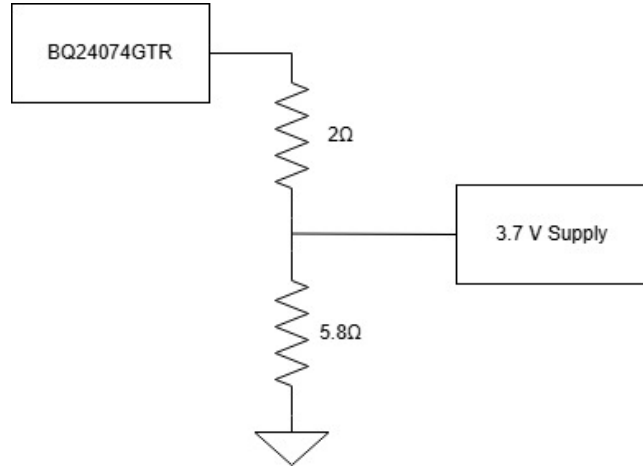


Figure 4.1: Dummy Battery for testing

The results align with findings in other tests, indicating the charging IC is likely in pre-charge mode due to either incorrect battery condition detection or temperature sensing issues. Further investigation into the charging circuit design is necessary, particularly focusing on the temperature sensing configuration.

4.3 AT03: Standard Charging Current

The objective of AT03 was to verify if the output current in standard charge mode was 200 mA. The test procedure involved connecting a dummy battery with a resistor banks and a 3.7 V source connected from a DC power supply, shown in the circuit diagram below.

The 2Ω and 5.8Ω are connected to the positive and negative terminals of the power supply, respectively. With the charger disconnected the current flowing through the 5.8Ω is $3.7\text{ V}/5.8\Omega = 638\text{ mA}$. After connecting the charger, current being supplied by the power supply should drop, with the difference being the charging current supplied by the BQ24074RGT .

Measured current from power supply with charger disconnected: 643 mA.

Measured current from power supply with charger connected: 643 mA.

Calculated current supplied by the BQ24074RGT : 0 mA

The result of this test is that the BQ24074RGT does not supply the required current to charge the battery. Upon further inspection into why this is the case, it was found that the schematic has an incorrect connection, in that pins 4, 6, and 14 of the BQ24074RGT had mistakenly been left as a floating point instead of grounded. A piece of wire was then soldered to remedy this, connecting TP6, which was connected to the affected pins, to TP8, which was at ground. The test was conducted again, but without a notable change in results.

The outcome of this test may be a result of the BQ24074RGT having an integrated temperature fault detection system, which was not considered during the design phase. The datasheet recommends ground the TS pin, which handles temperature faults and is usually connected to the battery, when not in use. This was done, but it seems to be causing issues and making the BQ24074RGT think that there is a temperature issue and should halt charging.

4.4 AT04: Fast Charging Current

The objective of AT04 was to verify if the output current in fast charge mode was 600 mA. The test was conducted in the exact same way as AT03, measuring the difference in current supplied by the power supply before and after a charger is connected, just instead having a voltage applied to FAST_CHARGE_CTRL to trigger the change in resistor network that biases the BQ24074RGT into supplying 600 mA.

Measured current from power supply with charger disconnected: 643 mA.

Measured current from power supply with charger connected: 645 mA.

Calculated current supplied by the BQ24074RGT : 2 mA

The result is the same as before, with very little current being supplied by the charging IC. Upon inspection, it was found that the charging IC would only output 2.2 V to charge the battery. It was also recognised that in the design phase, the IC charges the battery through the same lines that other components are connected to, which may cause them to become powered and create loading effects, which may cause the IC to enter power path management mode and divert power to these other components instead of charging the battery.

4.5 AT05: USB-C 9 V Output Verification

The objective of AT05 is to confirm if the output of the USB-C reaches 9 V.

The result of the test is a pass. The output of the USB-C reached 8.87 V and remained stable at this level, falling within acceptable tolerances outlined by the pass/fail criteria. The result confirms that the USB PD IC correctly negotiates approximately 9 V from a USB PD compliant power source. Additionally, the stability of the output further confirms the reliability of the USB PD negotiation circuit.

The successful result of AT05 is significant in that it verifies an important power interface for the entire subsystem. The USB-C serves as the main input for battery charging, its proper functioning ensures that downstream components such as the BQ24074RGT charging IC receives the correct supply voltage. The verified 9 V rules out power source issues as a cause for failures downstream, allowing focus to shift toward the configuration of those components.

Chapter 5

Conclusion

The acceptance testing results for the BQ24074RGT charging circuit reveal significant issues that prevent the system from functioning according to specifications. From these tests, several clear interpretations and conclusions can be drawn.

Firstly, voltages and currents measured in AT01 to AT04 indicate that the BQ24074RGT operates in pre-charge mode, as shown by the low output voltage and reduced current, well below the expected 200 mA and 600 mA. The minimal difference between standard and fast charge modes supports this. Secondly, AT05 confirms the USB-C PD system delivers a stable 8.87 V input, isolating the fault to the BQ24074RGT circuit rather than the power supply. Lastly, investigations revealed two main causes for this failure:

1. **Temperature sensor configuration:** The BQ24074RGT appears to be falsely detecting a temperature fault condition, likely due to improper configuration of the IC's TS pin.
2. **Power path management interference:** The charging path shares connections with other components, potentially causing the IC to divert power away from battery charging to system loads.

These findings directly address the initial problem: charging the battery. The BQ24074RGT, while operational, fails to meet the specified charging current requirements due to incorrect circuit design and configuration.

5.1 Recommendations

Based on these results, it is recommended that the following actions are taken for further development:

1. **Temperature sensing circuit redesign:** The TS pin configuration should be meticulously reviewed against the datasheet recommendations. A potential solution involves implementing a proper NTC thermistor network or ensuring proper pull-up/pull-down resistors are in place to prevent false temperature fault detection.
2. **Circuit isolation:** Redesign the power path to separate the battery charging circuit from other system components to prevent unwanted loading effects that may trigger the power path management mode.

In summary, while the current implementation is functional at a basic level, major circuit redesigns are necessary to meet the specified charging performance requirements and ensure reliable battery charging operation.

Chapter 6

Reflection

1. During the design phase, a more thorough review of the BQ24074RGT's datasheet and a simulation of its operation should have been conducted to flag potential issues. The layout of the final PCB could have been organised in a more efficient way and included more test points, pull-up/pull-down resistors and power paths could have been optimised for lower impedances. Earlier generation of fabrication files, followed by ERC/DRC checks would have minimized layout revisions. Clearly marking voltage nets and current paths in silkscreen would have also aided in debugging. Initial testing was performed without adequate instrumentation. Using dummy loads and step-by-step subsystem validation before full integration would have isolated faults more effectively.

2. Gantt



Figure 6.1: Gantt Chart of project lifecycle

3. The cost of the design could have been brought down by communicating more clearly with partner and organising the use of the same parts for non-critical components.

4. The stronger motor should be adjusted to match the output of the weaker motor using PWM from the processor. A gearbox could be implemented to match the rotation speeds of each wheel the motors are powering.

5. Shorts or stalled motors could lead to high current draw, damaging components. This can be avoided by implementing current limiting circuitry or thermal dissipation measures.